Eshan Ray

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	9.00
Disha Delphi Public School	Class XII (CBSE)	2024	94.60%
B.D.M.International	Class X (CBSE)	2022	98.00%

ACHIEVEMENTS

Gameathon 1.0 (FIFS)

- Secured 2nd place at Gameathon 1.0 (FIFS), a national-level inter-collegiate hackathon drawing teams from premier engineering institutes across India; awarded Rs. 7.5 Lakhs for developing an AI-driven cricket player performance prediction model with a custom weighted scoring algorithm that improved fantasy team selection accuracy by 15% over a baseline model.

Optiver Tradeathon

- Ranked 44th overall at the Optiver Tradeathon, a quantitative trading simulation competition.

PROJECTS

Microphone Amplifier with Class AB Driver -- EDC Lab Coursework | Dec-Jan 2026

- Designed a two-stage audio amplifier system comprising a low-noise BJT pre-amplifier (BC549, common-emitter with negative feedback) cascaded with a Class AB push-pull power stage (TIP31A/TIP32A) capable of driving an 8 speaker load.
- Achieved a voltage gain of 40 dB with a bandwidth extending into the 2-3 MHz range, verified through frequency response measurements on a Tektronix oscilloscope.
- Measured a Total Harmonic Distortion (THD) of 3.2% and an SNR of ~33 dB, confirming low-distortion operation suitable for audio amplification.
- Characterized key amplifier performance metrics including PSRR (~35.5 dB), quiescent current (20 mA), and maximum undistorted output power (0.8 W into 8).

LoRa-based Walkie-Talkie System -- PCB Design Coursework | Jan - Apr 2026

- Designed the complete PCB layout for a LoRa-based real-time voice communication system from schematic to fabrication using Altium Designer.
- Implemented audio encoding and decoding pipelines for low-latency voice transmission over the LoRa physical layer.
- Optimised antenna placement, impedance matching, and power delivery on the PCB to maximise transmission range and signal integrity.

Scientific Calculator Architecture -- Scientific Programming Coursework | Feb - Mar 2025

- Built a scientific calculator on Arduino (C++) with a hand-wired 5×5 matrix keypad on breadboard for input and an LED display for output, including shift/alpha keys that remapped button functions like a real scientific calculator.
- Implemented numerical methods (numerical integration, custom function approximations) and keypad-scanning logic in firmware, verifying computational accuracy against software reference implementations.

CNN Image Classifier with Knowledge Distillation -- Personal Project | Python, PyTorch | **GitHub**

- TDesigned and trained a convolutional neural network (3 conv blocks with batch normalization, max-pooling, and a fully connected classification head) on CIFAR-10 from scratch, reaching 77.7% test accuracy over 15 epochs using cross-entropy loss and the Adam optimizer.
- Built a 9.2× smaller student CNN (67,690 vs. 620,810 parameters) and trained it via knowledge distillation -- a temperature-scaled, combined loss of cross-entropy on true labels and KL-divergence against the teacher's softened output distribution -- rather than training on labels alone.
- Benchmarked both models on test accuracy, parameter count, and measured inference latency: the compressed student retained 85.1% of the teacher's accuracy (66.1% vs. 77.7%) while achieving a 1.6× faster inference time

(0.222ms vs. 0.363ms per sample), quantifying the real accuracy-efficiency tradeoff of model compression.

RELEVANT COURSES

Electrical Courses:- Digital Systems, Analog Devices, Control Systems, Signal Processing, PCB Design

CS and ML Courses:- Computer Architecture, DSA, Machine Learning

Math Courses:- Probability and Random Processes, Intro to Statistics, Complex Variables

SKILLS

Languages: C, C++, Python, Verilog, RISC-V Assembly

ML/DL Tools: PyTorch, torchvision, Scikit-learn, Pandas

Developer Tools: Git, GitHub, Linux (Ubuntu), Xilinx Vivado, Intel Quartus, GTKWave, Matplotlib, LaTeX

Relevant Foundations: Machine Learning, Model Compression & Distillation, Statistics & Probability, Signal Processing, Experimental Design & Performance Analysis

EXTRACURRICULAR

Matiks -- competitive math, logic & puzzle platform

- Achieved a 1600+ rating in Math, Logic, and Puzzles on Matiks, with a 2150+ rating specifically in Cross Math.

Codeforces Max rating **1205**

Sports & other interests: Badminton enthusiast (play regularly with friends); also enjoy swimming, table tennis and chess; Green belt in Karate.